



Module introduction

SMM921: Trading and Microstructure

Professor Richard Payne

Bayes Term 3 elective

[bayes.city.ac.uk]

$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$

What is this course about?

- **Part 1:** structure of modern financial markets
 - How modern (equity) markets work
 - Market players and market organisation
 - How are trading costs determined?
 - Electronic trading
 - Crypto assets and decentralised exchanges
- **Part 2:** quantitative portfolio management
 - Inputs: return, risk and trading cost forecasts
 - Building a quant model
 - Portfolio construction and evaluation

Thus there will be some theory, some econometrics/statistics, some examination of data and some coding. All analysis will be done in Python and code provided.

Organisation

Delivery is in 2 components

- **Live lectures:** 6, two-hour lectures, blocked in weeks 1 and 2 of term 3.
- **Online material:** recordings and quizzes to accompany each lecture. Usually applications of the lecture material that's relevant to the coursework.

Assessment

Group coursework: groups will be assigned a topic on which to write a report.

- You must self select into groups of at most 4 people.
- Topics will be assigned by course leader.
- The coursework will require you to do some quant analysis.
- Peer assessment operates as normal.

Reading

The three most important books;

- Thierry Foucault, Marco Pagano. and Ailsa Röell, *Market liquidity: theory, evidence, and policy*, Oxford University Press, 2023. [[ebook](#)]
- Ang (2014), *Asset Management*, OUP. [[ebook](#)]
- Grinold and Kahn (2000), *Active Portfolio Management*, McGraw-Hill. [[library record](#)]

Will supplement these with reading (placed on Moodle) from journals and other texts, as trading changes quickly so textbooks quickly become outdated.

Liquidity and portfolio management: linking parts 1 and 2

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$$\text{Utility} = \text{Mean return} - \text{Risk penalty} - \text{Costs}$$

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$$\mathcal{U} = w' \mu - \frac{\lambda}{2} w' \Sigma w - \tau' |w - w_0|$$

where

$$\tilde{1}' w = 1, \quad \tilde{1}' w_0 = 1$$

abs value bc all transactions incur costs
tau is how much it costs to trade that instrument on an instrument by instrument basis
embedding costs into the strategy
its not just an accounting cost at the end but it changes the allocation ex ante
this makes the optimization harder bc its not just a quadratic form
you have to do a numerical search

and w and w_0 are initial and final portfolio weight vectors, μ is a vector of expected returns, Σ is a return covariance matrix and τ is a vector of trading costs. λ is risk aversion. There are N assets.

Liquidity and portfolio management: linking parts 1 and 2

We will;

- Be concerned with understanding the determination of τ in Part 1

$$\text{Utility} = \text{Mean return} - \text{Risk penalty} - \underbrace{\text{Costs}}_{\text{Part1}} \quad (1)$$

- Be concerned with maximising the overall problem in Part 2

$$\underbrace{\text{Utility} = \text{Mean return} - \text{Risk penalty} - \text{Costs}}_{\text{Part2}} \quad (2)$$